

Plotting, Ordering and Comparing Real Numbers

Teacher's Edition

What Kind of Day This Is

This is a practice day, and it is built differently from the other five.

Math Nation's version of this lesson has no notes, no examples and no teacher-led instruction at all. It is three rotating stations plus a reflection, and the lesson narrative says the stations are “designed for students to be able to complete each station without needing guidance from the teacher.”

Rebuilt for a class with no group work, the shape here is: **a fast review of the unit's three skills from the front, three worked examples, and then a four-part circuit each student works alone.** The stations' mathematics is all preserved — every question from all three is on the circuit — but nobody rotates and nobody works in a group.

Because it is a practice day, the split is different: **40 minutes of instruction and 13 minutes of circuit**, the longest sheet time in the unit. The other five lessons run 41 to 45 minutes of instruction and 8 to 12 of worksheet.

Standard and Targets

MA.8.NSO.1.2 Plot, order and compare rational and irrational numbers, represented in various forms.

Boundary The expectation is to plot, order and compare square roots and cube roots, using $<$, $>$ and $=$.

Learning targets I can compare, order and plot every kind of real number in this unit.

Building on All five previous lessons. Nothing here is new.

Where this sits in the course Last content day of Unit 2. The Ticket Out the Door's third question asks students what they most want to look at again, which is the closest thing you will get to a free diagnostic before the assessment.

Materials Whiteboards and markers. Calculators open. Circuit 7A.02.06 — one sheet per student. No Frayer model templates and no station cards to print.

Lesson at a Glance

Windy Hill periods run **53 minutes**.

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up: same and different	2	4 min
Review I–III: check the radicand, convert then order, plot in two passes	3–6	9 min
Example 1: four comparisons, one equality	7–9	5 min
Example 2: rational or irrational, sorted and plotted	10–12	4 min

Segment	Slides	Time
Example 3: Sara's reasoning	13–15	5 min
Whiteboards: 9 questions, each with its reveal	16–33	20 min
IXL — start today's skills	34	5 min
Total		53 min

Where the lesson lives: Slides 13–15. Sara's error is the single most likely mistake on this benchmark and it is the last thing students see before the assessment.

No partner or group work anywhere in this lesson. Math Nation's version is three group stations with assigned partners inside each group. All of it is now individual: the review is teacher-led, the examples are worked from the front, and the circuit is worked alone at a desk.

Vocabulary

real numbers — the rationals and the irrationals together — everything on the number line

rational number — a number that can be written as a ratio of two integers

irrational number — a number that cannot be written as a ratio of two integers

index — the small number in the notch — it tells you which list of perfect powers to use

A note on the glossary. Math Nation lists none for this lesson. The four above are the ones the circuit uses, and “index” is repeated from Lesson 2 because slides 13–15 turns on it.

Slide-by-Slide Teaching Notes

Slide 1. Plotting, Ordering and Comparing Real Numbers *1 min*

- Post the learning target and say the framing honestly: there is no new content today.
- The first half is a fast review of the three things the unit asks for — sorting, comparing, plotting — plus the one error that costs the most.
- The second half is a four-part circuit students work alone.
- Tell them the assessment is next and that today is the rehearsal.

Slide 2. Warm-Up *4 min*

- TYPE: Same and different.
- INDEPENDENT.
- Expect: same — both are radicals, both are rational, both equal 6.
- Different — the index, and the radicand.
- The observation to fish for is the last one: they are THE SAME NUMBER, which is why '=' is one of the three choices on every comparison question and why students who never write it lose points. 216 is beyond the range the test asks about; say so, and say it is here because it is the next cube after 125 and worth knowing.
- OFF THE SLIDE, YOURS TO SAY: Two different-looking expressions. One number. That is worth two points on the test.

Slide 3. Review: I. Is it rational? Check the radicand, not the radical 3 min

- Fast review, not new notes — students should be filling this in from memory with you.
- Put both lists back on the board while you do it.
- Item C's second half is the one that gets missed: two thirds of π is irrational, because scaling by a non-zero rational number never repairs it.
- The words NON-ZERO are on the Reference Sheet and they are on this slide, and they earn their place — nought times π is nought, which is rational, and it is the only exception there is.
- OFF THE SLIDE, YOURS TO SAY: Perfect squares to 225. Perfect cubes to 125. Those two lists answer this.

Slide 4. Review: II. Convert first. Then order by position 3 min

- Item D is Lesson 5 in one line; ask a student to say it back before you move on.
- The panel sounds trivial and is not.
- Every ordering item in this unit asks LEAST TO GREATEST — review 34, 35 and 41, the study guide, the circuit, the 2.05 worksheet and additional practice questions 8 and 9, both Tickets Out the Door, the 2.06 additional practice, and assessment question 6 — so nothing here punishes autopilot; the habit is for the day something asks the other way.
- Do not tell them the test asks both directions.
- It does not, and a promise about the paper that the paper does not keep is worth less than the reading habit itself.
- OFF THE SLIDE, YOURS TO SAY: Read the question. “Least to greatest” and “greatest to least” are opposite instructions — check which one you were given.

Slide 5. Review: III. Plot: integers first, then the tenth 2 min

- Item A is the most common careless error on a plotting item: students assume every tick is 1 and plot six numbers wrongly in ten seconds.
- Read all four aloud; they are the checklist for the circuit.

Slide 6. Review 1 min

- The same six points they will meet in twenty minutes.
- Point at $\pi - 3$, which sits at about 0.14 and is the only one under 1, and at the three labels written below the line rather than above it.
- OFF THE SLIDE, YOURS TO SAY: Three labels sit below the line. That is rule D, and it is the only way six points fit.

Slide 7. Example 1 2 of the 5 min

- Whiteboards, decimals column only.
- Holding the symbol back until the decimals are written is the habit the whole unit is built on, and this is the last chance to enforce it before the test.
- Do not let anyone announce a symbol yet.
- OFF THE SLIDE, YOURS TO SAY: Fill the last column before you fill the middle one.

Slide 8. Example 1 1 of the 5 min

- Check the four rows of decimals before any symbol goes in.
- Row 3 is the one to circle: $\sqrt[3]{49} \approx 3.66$, so its opposite is -3.66 , which is nowhere near -7 .

- Ask for a show of hands on row 2 before you reveal it.
- OFF THE SLIDE, YOURS TO SAY: Row 2 has the same number on both sides. Now the symbols.

Slide 9. Example 1 *2 of the 5 min*

- All four rows are on the circuit.
- Row 1 is the negatives rule with an irrational on the left: -5π is about -15.71 , further left than -15 .
- Row 2 is the equality — a student who never writes $=$ will miss it, and that is why the warm-up was what it was.
- OFF THE SLIDE, YOURS TO SAY: Rows 1 and 3 are negatives. Row 2 looks different on both sides and is not.

Slide 10. Example 2 *2 of the 4 min*

- Whiteboards, middle row only.
- The one students get wrong is $\sqrt[3]{64} + 8$ — they see a radical and call it irrational, but $\sqrt[3]{64}$ is exactly 4, so the expression is exactly 12.
- Say it as a rule: work the radical out BEFORE you judge the expression.
- OFF THE SLIDE, YOURS TO SAY: One of these five has a radical in it and is a whole number. Find it before you sort anything.

Slide 11. Example 2 *1 of the 4 min*

- Do the middle row first, then the decimals.
- Note the panel out loud — the five happen to be listed in ascending order.
- So is every other ordering set on both sheets, the circuit included, so do not promise the class that the circuit will catch them.
- Ask for the DECIMAL VALUES as the evidence instead of the order: question 7 asks for a sort into rational and irrational, which the printed order does not give away, and questions 10 and 11 ask for a plot, which needs the values whether or not the order was free.
- OFF THE SLIDE, YOURS TO SAY: The bottom row is already in order. That is a coincidence today and not a method.

Slide 12. Example 2 *1 of the 4 min*

- This is the circuit's question 7.
- Point at 11.33 and 12: two points two thirds of a unit apart, which is where the label-above-and-below rule from Review III earns its place.
- OFF THE SLIDE, YOURS TO SAY: The plot follows from the decimals without any further thinking. That is the whole point of writing them down.

Slide 13. Example 3 *2 of the 5 min*

- Ask the right-hand question first and insist on it.
- Her METHOD is textbook: bound the number between two list entries, then decide which end it is nearer.
- Do not let anyone say 'she is wrong' and stop — a student who thinks Sara is simply careless will not check their own index either.
- This is Math Nation's own item, kept because it is excellent.

Slide 14. Example 3 *2 of the 5 min*

- Only the list is wrong.
- Put the two columns up side by side and let the class see that every line after the first is a correct consequence of a wrong start.
- That framing is what makes it a lesson about reading rather than a lesson about carelessness.
- OFF THE SLIDE, YOURS TO SAY: One word in the question — the index — and it changes both bounds.

Slide 15. Example 3 *1 of the 5 min*

- 3.3 against 2.2 is not a small miss, and pointing at the size of it is worth the ten seconds.
- Circuit question 9 is this item.
- OFF THE SLIDE, YOURS TO SAY: Read the index before you pick a list. That one habit is worth more points than any other in this unit.

Slide 16. Whiteboards · Question 1 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 17. Whiteboards · Question 1 of 9 — reveal *about 1 minute with the discussion*

- The baseline, and an equality first on purpose. 36 is a perfect square, so $\sqrt{36}$ is exactly 6 — not 'about 6 '.
- Everything after this changes exactly one thing.
- COLOR: none on this sequence — the reveal carries decimals rather than a coded pair, so under rule 5 there is no axis to name.

Slide 18. Whiteboards · Question 2 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 19. Whiteboards · Question 2 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the radicand, by one. $37 > 36$ so $\sqrt{37} > 6$, and no estimating is actually needed to see it.
- Expect boards that still say $=$ — that student compared the shapes, not the numbers.
- COLOR: none on this sequence — the reveal carries decimals rather than a coded pair, so under rule 5 there is no axis to name.

Slide 20. Whiteboards · Question 3 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 21. Whiteboards · Question 3 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: both signs.
- Identical numbers to question 2 and the answer flips. -6.08 is further left than -6.00 .

- This is Lesson 5's rule with one lesson of distance on it.
- COLOR: none on this sequence — the reveal carries decimals rather than a coded pair, so under rule 5 there is no axis to name.

Slide 22. Whiteboards · Question 4 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 23. Whiteboards · Question 4 of 9 — reveal *about 1 minute with the discussion*

- One feature changed from question 2: the index.
- Same 37, and now the answer is nowhere near $6 - 27 < 37 < 64$, so it is between 3 and 4.
- Sara's error is exactly the failure to notice this change.
- COLOR: none on this sequence — the reveal carries decimals rather than a coded pair, so under rule 5 there is no axis to name.

Slide 24. Whiteboards · Question 5 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 25. Whiteboards · Question 5 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the radicand, to the perfect cube.
- Back to an exact equality, and by a different route than question 1 — that one was a square, this one is a cube.
- COLOR: none on this sequence — the reveal carries decimals rather than a coded pair, so under rule 5 there is no axis to name.

Slide 26. Whiteboards · Question 6 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 27. Whiteboards · Question 6 of 9 — reveal *about 1 minute with the discussion*

- THE ONE THAT CLOSES THE UNIT.
- One feature changed: the right side is a radical too.
- Two expressions with different indices and different radicands, written with different symbols, and they are the same number.
- This is the warm-up returning as the last thing they do before the circuit, and it is the item the assessment asks.
- COLOR: none on this sequence — the reveal carries decimals rather than a coded pair, so under rule 5 there is no axis to name.

Slide 28. Whiteboards · Question 7 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.

- Do not reveal until every board is up.

Slide 29. Whiteboards · Question 7 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: NEITHER side is a whole number and neither is exact on a calculator display — the first comparison in the unit where both numbers have to be carried to several places before they separate. $22/7$ is 3.142857..., π is 3.141592..., so $22/7$ is GREATER.
- They differ in the third decimal place and nowhere earlier.
- THE BOARD TO LOOK FOR SAYS =, and it is the most understandable answer in the unit: the reference sheet says the Grade 8 FAST hands them $22/7$ to compute with, and a value you are given to calculate with is not the number. $22/7$ is rational and π is not, so they cannot be equal — which is a second, cleaner argument worth making after the decimals have done the work.
- COLOR: none on this sequence — the reveal carries decimals rather than a coded pair, so under rule 5 there is no axis to name.

Slide 30. Whiteboards · Question 8 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 31. Whiteboards · Question 8 of 9 — reveal *about 1 minute with the discussion*

- One feature changed from question 6: the right-hand side is irrational rather than another radical. $\sqrt[3]{216}$ is exactly 6 and π is 3.14..., so the left side is greater, and this one should take five seconds.
- It is here deliberately as the easy question after the hard one — question 7 separates in the third decimal place, and this separates in the units.
- Part of comparing numbers is recognizing when you do NOT need to be careful, and a round that only ever asks for precision teaches students to spend it everywhere.
- COLOR: none on this sequence — the reveal carries decimals rather than a coded pair, so under rule 5 there is no axis to name.

Slide 32. Whiteboards · Question 9 of 9 — ask *about 90 seconds of writing*

- Boards down until the cue, all up together.
- This one is written, so give it the time — a board going up early here is a board with one word on it.

Slide 33. Whiteboards · Question 9 of 9 — reveal *about 2 minutes with the closing discussion*

- WRITTEN WORK, and the only question in the round that asks for it.
- With the worksheet gone this is the only place in the period the habit gets built, and IXL takes an answer without ever seeing the work.
- ANSWER: $-\sqrt[3]{37}$, $\sqrt[3]{37}$, 6, $\sqrt{37}$ — that is -6.08, 3.33, 6.00 and 6.08.
- EVERY NUMBER HERE IS BUILT FROM 37 and they land in four different places, which is the whole unit in one line: the index changes the size, the minus sign changes the side, and 6 is the fixed point they are all measured against.
- THE ORDERING TO LOOK FOR PUTS $\sqrt[3]{37}$ LAST, from a student reading 37 and 3 and deciding a cube root of 37 must be large.
- It is 3.33, smaller than 6.

- THE MARK IS FOR THE METHOD — a student who writes four decimals down and then orders them has done exactly the right thing and should say so, because that is what the assessment wants to see.

Slide 34. IXL 5 min

- IXL.
- The skills are assigned in Google Classroom, not named here, and they are deliberately not ours -- Croix: "The students need questions that aren't exclusively mine."
- Makes them think." Circulate.
- The thing to watch for is a student answering in the box with no paper in front of them: IXL takes an answer and never sees the work, so this is the one part of the period where the habit can quietly lapse.

Circuit 7A.02.06 — Answer Key

PART 1	1) < 2) = 3) < 4) < 5) > 6) < Values: -15.708 vs -15 ; 5 vs 5 ; 6.708 vs 7.142 ; -7 vs -3.659 ; 9.667 vs 8.944 ; 1.047 vs 1.690 .
4. -7 vs $-\sqrt[3]{49}$	< The one to watch. $\sqrt[3]{49} \approx 3.66$, so its opposite is -3.66 , well to the right of -7 .
2. $\sqrt[3]{125}$ vs $\sqrt{100} \div 2$	= The equality. A student who never writes $=$ has not checked it.
7. sort and plot	Rational: 10.2, $11\frac{1}{3}$, $\sqrt[3]{64} + 8$ Irrational: $-\pi + 16$, $\sqrt{190}$ $\sqrt[3]{64} + 8 = 4 + 8 = 12$ exactly. Values: 10.2; 11.333; 12; 12.858; 13.784.
8. plot six negatives	-5.455; -5; -4.472; -3.5; -3.142; -2.828 All six are negative, so the one that looks biggest is furthest left.
9a. Sara's error	She used perfect squares to estimate a cube root. Her method is right. Only her list is wrong. Accept any wording that says so.
9b. done correctly	$8 < 11 < 27$, so $\sqrt[3]{11}$ is between 2 and 3, and closer to 2. $\sqrt[3]{11} \approx 2.2$ $11 - 8 = 3$ and $27 - 11 = 16$.
10. order and plot	$\sqrt[3]{8}$, π, $\sqrt[3]{100}$, $\sqrt{48}$, $19/2$, $\sqrt{121}$ Values: 2; 3.142; 4.642; 6.928; 9.5; 11. Two of the six are whole numbers in disguise.
11. order and plot	$\pi - 3$, $\sqrt[3]{49}$, $\sqrt[3]{125}$, $7\frac{1}{2}$, $\sqrt{80}$, $\sqrt{115}$ Values: 0.142; 3.659; 5; 7.5; 8.944; 10.724.
12. stretch	$\sqrt[3]{216} = 6$ $\sqrt[3]{-343} = -7$ Beyond the tested range and labeled as such on the sheet.

Part 3 is the first of the two to circulate for; question 12 is the other, below. A student who writes only “she used squares instead of cubes” has the answer. A student who also says what was right about Sara's method has understood the slide.

Question 12 is marked STRETCH on the sheet. It uses 216 and 343, beyond MA.8.AR.2.3's stated range of -125 to 125 . It is labeled in print so nobody — student or substitute — mistakes it for required fluency.

Additional Practice 7A.02.06 — Answer Key

PART 1	1) > 2) < 3) > 4) < 5) = 6) < Values: 3.717 vs -3.826; 17.321 vs 17.818; -9 vs -9.0909; 30 vs 31.006; 12 vs 12; 4.021 vs 4.25.
3. $-\sqrt{(162/2)}$ vs $-9.\overline{09}$	> $162 \div 2 = 81$, so the radical is exactly -9. Worth slowing down on.
7. sort and plot	Rational: $2\frac{1}{2}$, $\sqrt[3]{64}$, $19/3$ Irrational: $\sqrt{10}$, $\pi + 2$ $\sqrt[3]{64} = 4$ exactly — the circuit's $\sqrt[3]{64} + 8$ trap, one step shorter. Values: 2.5; 3.162; 4; 5.142; 6.333.
8. plot six negatives	-6.708; -6.333; -5.477; -5.196; -4; -3.742 $-\sqrt{30}$ and $-\sqrt{27}$ sit 0.281 apart, so one of those two labels goes below the line.
9a. Devin's error	He used perfect cubes to estimate a square root. Sara's error running the other way. The index tells you which list to use.
9b. done correctly	$36 < 40 < 49$, so $\sqrt{40}$ is between 6 and 7, and closer to 6. $\sqrt{40} \approx 6.3$ $40 - 36 = 4$ and $49 - 40 = 9$.
10. order and plot	$\sqrt[3]{1}$, $\sqrt[3]{27}$, $\sqrt{15}$, $\sqrt{40}$, $31/4$, π^2 Values: 1; 3; 3.873; 6.325; 7.75; 9.870. Two of the six are whole numbers in disguise.
11. order and plot	$1/2$, $\sqrt{8}$, $5\frac{1}{2}$, $\sqrt{55}$, 3π, $\sqrt{126}$ Values: 0.5; 2.828; 5.5; 7.416; 9.425; 11.225.
12. stretch	$\sqrt[3]{512} = 8$ $\sqrt[3]{-216} = -6$

This sheet is optional. Post it to Google Classroom. It is not assigned and it is not collected.

Changes from the Math Nation Lesson

- **The three stations became a four-part circuit worked alone.** Every question from all three stations is on the sheet. What is gone is the rotation, the grouping and the assigned partners inside each group — and, with them, the station cards and the Frayer model templates.
- **A review-and-examples half was added.** Their lesson has no instruction in it at all. Since this is the last day before the assessment and the class does not do group discovery, the first 40 minutes are now four review slides, three worked examples including the Sara slide, the warm-up, the whiteboard sequence and the ticket.
- **The worst ordering item in the unit was retuned.** Their practice sheet question 10 asks students to order $\sqrt[3]{-126} = -5.01330$, $-5.0\overline{1} = -5.01111$ and $-\sqrt{25} = -5.00000$ — a spread of 0.013 across all three — and then to plot the result. No estimation strategy reaches four decimal places. That item is gone; the ordering sets on this circuit have gaps of at least a tenth throughout.
- **Cube roots were brought back to the taught list.** Their sheets use $\sqrt[3]{530}$, which needs 512 and 729, and $\sqrt[3]{-212}$ and $\sqrt[3]{-126}$, which need 216. None of those cubes appears in any benchmark's stated range. The circuit's core items use cubes from the Lesson 3 list, and 216, 343 and 512 appear only in the STRETCH question, which says so in print.
- **Two questions that could not be answered by estimating were replaced.** Their homework question 9 asks for the order of $123/15 = 8.2000$, $\sqrt{70} = 8.3666$ and $3\pi - 1 = 8.4248$; their practice question 2 asks whether $-28/6 = -4.66667$ or $\sqrt[3]{-100} = -4.64159$ is greater. The three in question 9 sit 0.167 and 0.058 apart and the practice pair 0.025 apart — gaps no estimation method on this sheet can resolve, so both are calculator exercises rather than estimation ones.
- **Two good items were kept as they were.** The Bell Work warm-up and Sara's error analysis are both well built, on benchmark and individual as written. Sara's item is now the named-difficulty slide as well as a circuit question.
- The wrap-up was replaced. Theirs asks students to reflect on the unit learning targets. The Ticket Out the Door asks for an ordering, a justification, and one sentence naming what they want to revisit — which is the most useful thing you can collect on the day before an assessment.
- **Nothing else in the mathematics was changed.** Every comparison and every ordering on their sheets was checked, and apart from the retunings named above, every one of them was correct.